

To Implement Single Quote Handling for Literal String Preservation

```
LST@LAPTOP-EK5DCJVE:~# mkdir -p ~/quote_lab
LST@LAPTOP-EK5DCJVE:~# cd ~/quote_lab
LST@LAPTOP-EK5DCJVE:~/quote_lab# pwd
/root/quote_lab
LST@LAPTOP-EK5DCJVE:~/quote_lab# nano quote.c
LST@LAPTOP-EK5DCJVE:~/quote_lab# gcc quote.c -o quote
LST@LAPTOP-EK5DCJVE:~/quote_lab# ls
```

Code:

```
GNU nano 7.2
#include <stdio.h>
#include <string.h>

#define MAX 1000

void handle_single_quotes(char *input, char *output)
{
    int i = 0;
    int j = 0;
    int inside_single_quote = 0;

    while (input[i] != '\0')
    {
        if (input[i] == '\'' )
        {
            // Toggle single quote state
            inside_single_quote = !inside_single_quote;
        }
        else
        {
            // Copy character literally
            output[j++] = input[i];
        }

        i++;
    }

    output[j] = '\0';
}

int main()
{
    char input[MAX];
    char output[MAX];

    printf("=====\n");
    printf("SINGLE QUOTE HANDLING\n");
    printf("=====\n");

    printf("Enter a string: ");

    fgets(input, MAX, stdin);

    // Remove newline
    input[strcspn(input, "\n")] = '\0';

    handle_single_quotes(input, output);

    printf("\nOriginal string : %s\n", input);
    printf("Processed string: %s\n", output);

    return 0;
}
```

Execution:

```
LST@LAPTOP-EK5DCJVE:~/quote_lab# ./quote
=====
SINGLE QUOTE HANDLING
=====
Enter a string: 'Hello $USER'

Original string : 'Hello $USER'
Processed string: Hello $USER
LST@LAPTOP-EK5DCJVE:~/quote_lab# |
```

```
Processed string: Hello $USER
LST@LAPTOP-EK5DCJVE:~/quote_lab# ./quote
=====
SINGLE QUOTE HANDLING
=====
Enter a string: 'Hello * ? $HOME # @'

Original string : 'Hello * ? $HOME # @'
Processed string: Hello * ? $HOME # @
LST@LAPTOP-EK5DCJVE:~/quote_lab# |
```